

JAYCE(SEONGJIN) CHOI

MANUFACTURING SYSTEMS ENGINEER (MES / SMART FACTORY)

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PROFILE

Manufacturing Systems Engineer specializing in MES/MCS integration, smart factory deployment, and industrial automation for large-scale U.S. EV battery plants. Expert in leading equipment interface integration (CIM/MCS), SQL-driven material flow optimization (RTD/MHS), and deployment integrity automation. Proven track record of reducing verification time by 90% and ensuring 100% source-to-production consistency. Experienced in supporting 24/7 high-volume manufacturing across multi-site operations (OH, MI, GA).

PROFESSIONAL EXPERIENCE

Manufacturing Systems Engineer(MES), LG CNS America **Jan 2023 to Present**
Warren, OH / Holt, MI / Ellabell, GA

Played a key role in large-scale smart factory deployment for multi-site U.S. EV battery plants operating 24/7 high-throughput production environments.

- Led MES/MCS/CIM integration with automated material handling systems (conveyors, stacker cranes, AGVs/AMRs), enabling real-time material tracking across EV battery production lines.
- Architected and deployed a Python-based MES Deployment Diagnostic Framework using GitLab REST API and Regex validation logic, reducing pre-deployment verification time by 90%.
- Implemented automated 3-way reconciliation (GitLab source vs. Server API vs. Deployment list), ensuring 100% system integrity across BizActor and service IDs.
- Optimized RTD/MHS SQL logic to improve material routing accuracy, reduce dispatch latency, and enhance production stability.
- Conducted cross-layer system validation across MHS–WMS–ERP integrations to ensure end-to-end material traceability.
- Performed deep-dive root cause analysis of recurring MES/MCS production incidents within centralized IT shared service environments.
- Supported commissioning and on-site go-live stabilization for multiple U.S. battery plants operating in 24/7 high-throughput production environments.
- Collaborated with cross-functional teams including controls engineers, IT infrastructure, and production stakeholders to align automation logic with operational requirements.

EDUCATION

Hongik University **Mar 2014 to Feb 2018**
Bachelor of Computer Science, Seoul, South Korea

AREAS OF EXPERTISE

Manufacturing Systems Architecture (MES/MCS/CIM, RTD/MHS)	Oracle SQL Performance Engineering	Industrial Automation Protocols (PLC, OPC-UA, HSMS-GEM)	Event-Driven Distributed Systems (Solace, Pub/Sub Messaging)
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TECHNICAL PROJECT

MES Deployment & Source Integrity Diagnostic Tool

- Developed an automation solution using **Python** and **GitLab REST API** to validate consistency between source code changes and production server instances.
- Implemented a **3-way reconciliation logic** (GitLab vs. API vs. Deployment List) using **Regex** to ensure 100% integrity of BizActor and Service IDs.
- Achieved a **90% reduction** in manual verification time and prevented potential service omissions across multi-site battery plants.

Python Logistics Visualization Tool

- Developed a Python-based visualization tool to analyze logistics flow, routing patterns, equipment anomalies, and process performance using real MES/MCS/RTD data. Converted raw SQL outputs into clear graphical insights, enabling faster troubleshooting and data-driven decision-making for smart factory operations.